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GPTutor: A Generative AI-powered Intelligent Tutoring System to Support Interactive Learning with Knowledge-Grounded Question Answering

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***Abstract*—With increasing popularity of artificial intelligence (AI) in the education industry, intelligent tutoring system (ITS) powered by AI have been widely adopted to optimize the learning experience. However, the relationship between students' engagement level of Generative AI (GenAI) and their academic performance is still under exploration. Also, current popular GenAI products like ChatGPT suffer from the hallucination problem, which includes factuality, faithfulness, and maliciousness issues in the generated answer. This paper presents GPTutor, an ITS leveraging GenAI to support students' learning processes. GPTutor integrates a Retrieval-Augmented Generation (RAG) pipeline to deliver actual and contextually rich answers aligned to student questions and intended learning outcomes (ILO). A pilot evaluation involving undergraduate and postgraduate students assessed the system's association with user experience, engagement, and academic performance. Results demonstrated that students generally recognize the effectiveness of GPTutor. Some students also provided insightful feedback on the benefits of GPTutor in improving learning efficiency and some limitations to be addressed. Notably, students with higher engagement levels showed significantly better academic performance on the final exam. This study proposed GPTutor to provide an interactive and knowledge-grounded learning experience and showed the strong association between students' engagement in GPTutor and academic performance.**

Keywords—Intelligent Tutoring System, Generative AI, Interactive Learning, Knowledge-Grounded Question-Answering, Retrieval-Augmented Generation

I. INTRODUCTION

Intelligent Tutoring Systems (ITS) have significantly advanced the field of education by providing adaptive and personalized learning experiences. These systems utilize artificial intelligence (AI) to dynamically adjust the difficulty of learning tasks, offer tailored guidance, and present educational content based on the learner's performance and preferences. Studies have demonstrated that ITS can effectively mimic one-on-one tutoring, often resulting in improved student performance and increased mastery of subjects [1]. Furthermore, ITS have been shown to enhance student engagement and motivation, contributing to deeper learning and retention of knowledge [2]. Generative AI (GenAI), a subset of AI that focuses on creating new content from existing data, has immense potential to revolutionize how students access information and learn innovatively. GenAI has been integrated into different intelligent tutoring systems. For instance, CS50 Duck [3], a GenAI-powered virtual AI assistant developed by Stanford University, offers an interactive environment that helps learners develop programming skills through practice and immediate feedback. Moreover, GenAI-powered ITS has been applied in professional training contexts, such as healthcare, where they assist medical students in acquiring clinical reasoning skills and decision-making through simulated patient interactions [4, 5]. Despite the increasing popularity of GenAI, its effectiveness in improving students' academic performance and engagement is still unclear. One mixed methods study, for example, showed that educators had no clear consensus on the impact of GenAI in higher education and expressed uncertainty about how to best adapt to the rapid development of GenAI. They also raised concerns about students' lack of engagement with information due to its convenient accessibility through GenAI [6]. Thus, this study designed a GenAI-powered ITS and investigated its effect on student engagement and academic performance in the empirical study.

This study developed GPTutor, a GenAI platform to empower subject-specific ITS that aims to deepen students' understanding of the course's contents by providing interactive exploration of learning materials. GPTutor allows instructors to create courses, and upload learning resources (e.g., lecture slides). GenAI is used to summarize the learning contents, generate review exercises, and provide a conversation interface for question-answering on learning resources. This study also integrated Retrieval-augmented Generation (RAG) into GPTutor to provide knowledge-grounded question-answering (KGQA) by using instructors' learning materials as a knowledge base to ensure that the answers provided are factual and reliable and provide better alignment to the course's learning outcomes. An instructor dashboard with learning activities is provided to allow instructors to monitor students' learning performance and engagement with the platform.

To test the effectiveness of GPTutor on student engagement and promoting academic performance, a pilot evaluation study was conducted in two courses related to software engineering/development at The Hong Kong Polytechnic University in Semester 2, 2023-2024 Students were invited to complete a survey, after using GPTutor to learn and review for a quiz. The students' perceived usefulness, attitude towards using GPTutor in learning, and their intention to use the GenAI-powered ITS platform for

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learning and revision were studied. Additionally, this paper analyzed students' usage behavior on the ITS platform to study students' engagement and its potential association with students' academic performance.

This paper is organized as follows. Section II presents a literature review about ITS integrated with GenAI and related challenges of unreliable answers and possible solutions. Also, Section II discusses the potential relationship between the usage of GPTutor and students' academic performance. Section III introduces the design and features of the product, GPTutor. Section IV evaluates the effectiveness of GPTutor. Finally, Section V concludes this paper and discusses the directions for future research.

II. Literature Review

A. Intelligent Tutoring System Powered by Artificial Intelligence

An ITS is a computer system that assists the teaching and learning process with four modules: expert module, student diagnosis module, tutor module, and user interface module [7]. With the emergence of AI technologies and their application in the education field, ITS has adopted AI to enhance its functionality [8]. Based on students' specific learning characteristics and habits, GenAI-powered ITS provides adaptive feedback to students and customizes their learning experience with the integration of natural language processing (NLP) [9], fuzzy-based methods [10], Bayesian networks [11], and data mining [12]. After the release of ChatGPT [13], educational platforms like CS50 Duck for programming guidance [3], Geeky Medics for simulated clinical chat on medical education [14], and Q-Chat for the second language vocabulary learning [15] explored integrating GenAI models with traditional teaching methods as a promising new paradigm for enhancing students' learning experiences.

B. Generative AI and Academic Performance in Higher Education

Although many educational products have been integrated with GenAI models or features [3, 4, 5, 14, 15], the impact of using GenAI in the learning process on academic performance is still under-explored. The research about GenAI integration on academic performance proposed that using GenAI can boost students' motivation and ultimately improve their academic performance [16]. Regarding the learning outcomes, research on a software modeling class indicated that computer science students, who participated in the ChatGPT formative assessment, performed better in the summative assessment compared with students who only received traditional teaching [17]. However, other researchers have raised concerns about using GenAI in higher education. They stated that students who have excessively used ChatGPT were more likely to procrastinate and have memory loss, eventually decreasing their academic performance [18].

C. Challenges of Involving Generative AI in Learning

Despite the popularity of GenAI among students, incorporating GenAI in the learning process presents challenges for educators. One research study on the generational differences in attitudes toward GenAI found that Generation Z students were more optimistic about the integration of GenAI in learning, in comparison with Generation X or Generation Y teachers [19]. However, both students and teachers raised concerns about potential challenges of misinformation generated by GenAI. Therefore, the responsible usage of reliable and high-quality content should be promoted in the ITS and other educational systems incorporating GenAI. Besides the perception of students and teachers, the response generated by GenAI tools like ChatGPT is unreliable [20]. Large Language Models (LLMs), such as ChatGPT, often produce content that, while plausible and natural, is factually incorrect. This issue, known as *factuality*, arises when the generated content contradicts universal facts (e.g., the sun rising in the East). The lack of factuality in the ChatGPT response may not be beneficial for students in acquiring knowledge and preparing for their future careers. Additionally, generated content may deviate from the input context, a problem referred to as *faithfulness* [21]. The lack of faithfulness may prevent students from following the intended learning outcomes (ILO) set by teachers and negatively impact their assessment results, even if they stress the ILO in the prompts of ChatGPT. Also, the response of GenAI could be malicious because of the veracity of training data and improper prompts. Specifically, prompt injection and jailbreak attacks make it possible to leak the offensive content to students [22]. This problem is referred to as *maliciousness*.

D. Knowledge-Ground Question-Answering

To mitigate the issues of *factuality*, *faithfulness*, and *maliciousness*, this paper proposed integrating Retrieval-Augmented Generation (RAG) into ITS to provide the Knowledge-Grounded Question-Answering (KGQA) service, as one of the core features of GPTutor. RAG is widely adopted to improve the factuality of the generated content by retrieving relevant chunks from the knowledge base and synthesizing coherent answers [23]. For the KGQA purpose, the common retrieval sources include encyclopedic data [24, 25], datasets [26, 27], and search engines [28, 29].

III. Design and Features of GPTutor

A. Intelligent Tutoring System Modules and System Architecture

As an ITS powered by GenAI, GPTutor incorporates the four classical modules of ITS [7]. The *expert module* serves as the knowledge base, storing the learning materials uploaded by instructors. Based on learning materials in the expert module, the *tutor module* can offer concrete and adaptive responses to students' questions via KGQA. To facilitate the instructors' understanding of students' learning progress and habits, the *student diagnosis module* serves as the dashboard to present useful statistics, including the leaderboard and conversations of KGQA. Finally, the *user interface module* supports the natural and convenient interactions between the users and the GPTutor system. Fig. 1 and Fig. 2 show the user interfaces for students (KGQA feature) and instructors (students' dashboard), respectively. Fig. 3 shows the system architecture of GPTutor. The Web Server interacts with the API Server via GraphQL [30]. To implement KGQA, an end-to-end RAG pipeline is designed and implemented. To support the specific storage requirements of this RAG pipeline, the system architecture includes an object store for storing the instructor's learning

contents and a vector store for indexing high-dimensional representations of the learning materials.

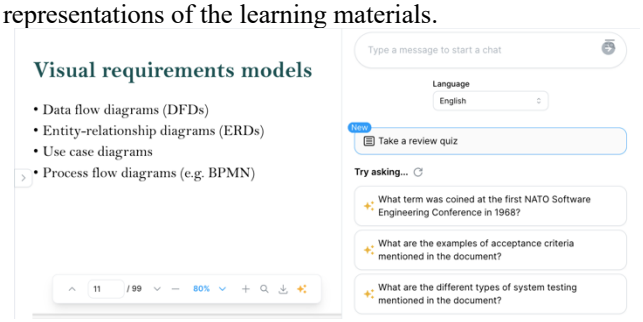

Fig. 1. A screenshot showing the students' user interface.

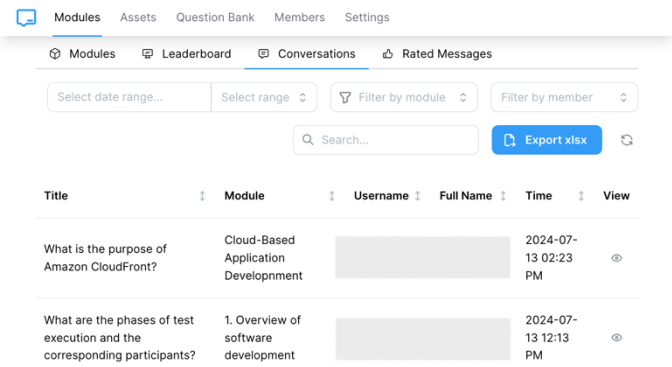

Fig. 2. A screenshot showing the instructors' user interface.

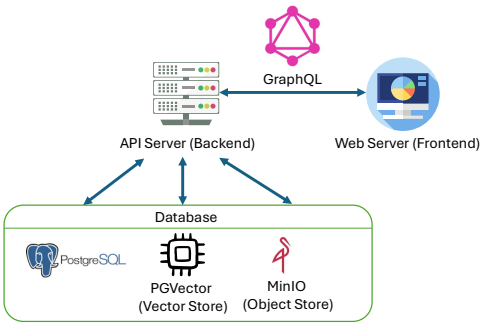

Fig. 3. The system architecture diagram of GPTutor.

B. *Retrieval-Augmented Generation System Implementation*

This study used the RAG pipeline to implement the core KGQA service, offering students a factual, faithful, up-to-date, and non-malicious chat-based learning experience. Following the classic RAG paradigm, the RAG pipeline consists of three main steps: indexing, retrieval, and generation. Fig. 4 and Fig. 5 show the implementation of the RAG pipeline. Before the three main steps, given the uploaded learning materials, the preprocessing will generate the document class and summary to enable students to overview the materials quickly. The indexing step incorporates the customized token-based splitter and embedding model. The splitter will split the documents into text chunks for storage. In RAG, using smaller chunks can create embeddings that better capture the meaning of each chunk to enhance the precision of information retrieval. However, each chunk might not contain enough context to generate comprehensive answers [23]. On the other hand, larger chunks provide more comprehensive information for answer generation, but they may increase the cost of preprocessing and storage, as well as introduce noise in retrieval results [23]. To address these challenges, this study adopted the concept of small-to-big retrieval [23] to create two-level indexing. The first page-wise splitting, based on the token size and overlapping size, will create the parent (bigger) chunks. After splitting the parent chunks, metadata will be added to each chunk, including a chunk identifier and a page identifier. This metadata will be used as the filter criteria during the retrieval step. Child chunks are created by further splitting the parent chunks and are assigned the corresponding parent chunk identifier. All chunks, along with their embedding vectors generated by the embedding models, are stored in the vector store.

After the indexing step, the RAG pipeline is ready for question-answering, which involves the retrieval and generation steps. When the user inputs a message, this question will go through the preprocessing steps, such as determining the purpose of the student's query using intention detection and rephrasing the student's question to provide more context for better retrieval in the RAG pipeline. If the intent detection indicates the question is irrelevant to the topic, the completion model will generate a standardized response, bypassing the RAG process. The same embedding model as the one used for chunk embedding will generate the embedding vector for the rephrased question. Using similarity metrics on the embedding vectors of the question and child chunks, the top-k relevant small chunks exceeding the similarity threshold are retrieved from the vector database, while the corresponding large chunks are used to provide more context during response duration. If no eligible child chunk is found in the retrieval step, the completion model will either answer the question based on its internal knowledge or decline to answer, depending on whether its internal knowledge suffices. Eventually, the completion model generates a coherent answer and reference links by combining the parent chunk, original question, intent class, and chat history.

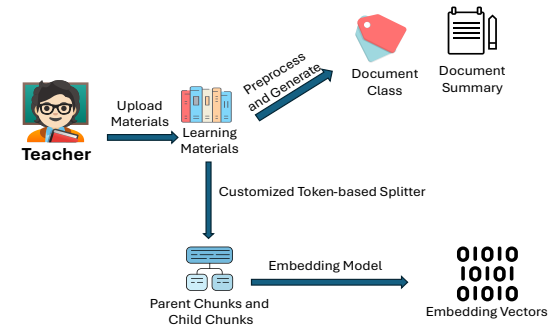

Fig. 4. Indexing steps of the RAG Pipeline.

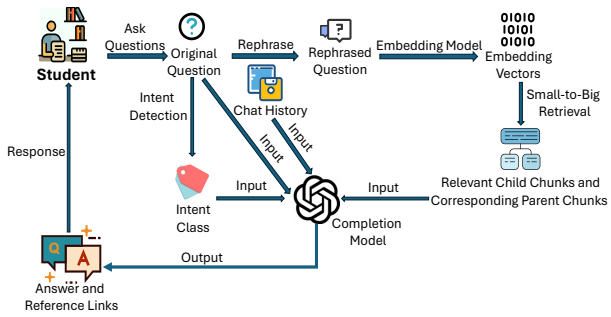

Fig. 5. Retrieval and Generation steps of the RAG Pipeline.

C. Features

Fig. 6 shows how users may interact with GPTutor. To set up the GPTutor system, the administrator will pre-define the system configurations, including the parameters of the splitter and retriever, the API specification of embedding and the completion model, and the system prompt templates. The administrator also monitors the dashboard of the Langfuse traces, which serves as the log of all conversation history and cost. The instructors can create and set up the courses. The course setup includes adding students to the course, uploading learning materials, creating the learning modules to enable the KGQA chatbot supported by the selected materials, and generating review exercises. By reviewing the leaderboard and conversation history, the instructors can review the correctness of AI-generated answers and the students' engagement in the platform.

After the students are added to the course, the student can select a learning module and interact with the KGQA chatbot. For each learning module, starter questions are provided to help students initiate conversations and explore the learning materials, even if they are unfamiliar with the topic. The AI-generated response will include reference links and follow-up questions. The reference links are generated based on the most relevant pages to the question from lecture notes during the retrieval process. Students can click the reference links to review the corresponding lecture notes and validate the factuality of the generated content. Meanwhile, the follow-up questions are designed to facilitate students exploring related questions or engaging in a deeper discussion of the topics. The students can also rate the AI-generated responses as the feedback to improve the accuracy and relevance of future responses.

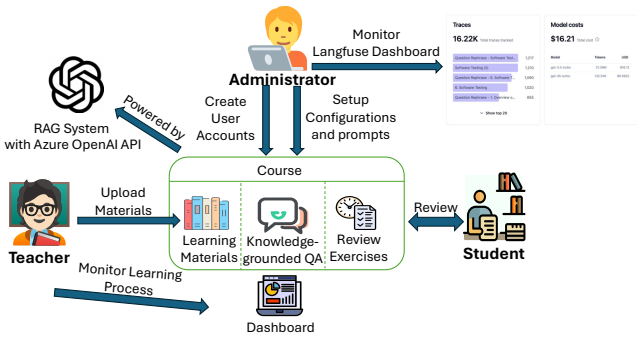

Fig. 6. The User-GPTutor interaction diagram.

IV. Evaluation

A. Pilot Study

To evaluate the effectiveness of GPTutor, a pilot study was conducted with 73 postgraduate students and 56 undergraduate students enrolled in two software engineering courses at The Hong Kong Polytechnic University. With lecture contents uploaded to GPTutor, students were invited to use the platform for learning and revision for around four weeks. This paper conducted two pilot studies. Study 1 involved administering a survey to investigate the students' usage experience. This included collecting qualitative feedback and measuring quantitative constructs, including perceived ease of use (EU), perceived usefulness (PU), intention to use the system (IU), and enjoyment (EJ). The survey items are adapted from the previous work, which interpreted these four constructs as a quantitative survey [31]. In Study 2, the learning analytics in the ITS platform was collected to investigate students' engagement and its relationship with their academic performance in the course.

For Study 1, the quantitative constructs of interest (measured on a 7-point Likert scale, ranging from 1 = "Strongly Disagree" to 7 = "Strongly Agree") are shown in TABLE 1. Overall, 60 postgraduate students and 36 undergraduate students completed the survey. The independent samples two-tailed T-tests were performed to analyze the difference in mean scores of these constructs between undergraduate students and postgraduate students. For collecting qualitative feedback, the following questions were included in the survey: "Please share your thoughts about whether the answer provided by GPTutor is meeting your expectation. Provide examples if available", "Do you think the platform is useful for learning in this course? Do you have any suggestion on what features we can add in GPTutor to increase the students' engagement in the platform?", and "What do you like or dislike about the platform? Any other suggestion for future improvement?" In Study 2, based on their level of engagement with GPTutor, the students were divided into two groups: the low-engagement group (i.e., students sending less than 10 messages) and the high-engagement group (i.e., students sending 10 or more messages). The subsample sizes were balanced, with 66 students in the low-engagement group and 63 students in the high-engagement group. The independent samples two-tailed T-tests were performed to analyze the difference in mean exam scores of these two groups.

TABLE 1. Survey items of Study 1

Construct	Items
Perceived Ease of Use of the Platform (EU)	EU1. My interaction with GPTutor is clear and understandable.
	EU2. Interacting with GPTutor does not require a lot of my mental effort.
	EU3. I find GPTutor to be easy to use.
	EU4. I find it easy to get GPTutor to do what I want it to do.
Perceived Usefulness of the Platform (PU)	PU1. Using GPTutor improves my performance in my course.
	PU2. Using GPTutor in my learning improves my productivity.
	PU3. Using GPTutor enhances my effectiveness in my learning.
	PU4. I find GPTutor to be useful in my learning.
Intention to Use (IU)	IU1. I will recommend GPTutor to others.
	IU2. I intend to continue using GPTutor after completion of the course.
	IU3. I expect my use of GPTutor to continue in the future after the completion of the course.
	IU4. Assuming I had access to GPTutor, I intend to use it.
Enjoyment (EJ)	EJ1. I find using GPTutor to be enjoyable.
	EJ2. The process of using GpTutor is pleasant.
	EJ3. I have fun using GPTutor.
	EJ4. I find interacting with GPTutor to be an engaging experience.

B. Results and Discussion

TABLE 2 shows the aggregate scores of the key variables of interest for the total sample. In general, the results suggest that GPTutor had a positive reception and was well used by the students. As shown in TABLE 3, the average survey scores of undergraduates on EU, PU, and EJ were found to be higher than postgraduates, while postgraduates had a lower mean score on IU. However, none of the two-tailed T-tests on these four metrics showed a statistically significant difference at the *p*-value threshold of 0.05. This suggests that undergraduate students and postgraduate students did not significantly differ in the usage experience of GPTutor.

Regarding the qualitative feedback, many students found the platform to be particularly useful for providing relevant answers that are well-aligned with the lecture notes. The chatbot's answers also included reference page numbers to help students quickly locate the information in the lecture notes. Additionally, students appreciated the ability to view the lecture notes alongside the chatbot, which enhanced their study experience. The suggested follow-up questions were helpful in assisting students achieve a deeper understanding of the subject content. A part-time postgraduate student appreciated GPTutor's ability to extract key points from the lecture slides, which facilitated an efficient review of the lecture content under a tight schedule. For limitations, some postgraduate students mentioned that the GPTutor tends to repeat content from lecture slides, rather than offer new insights, and they desired more in-depth discussions that extended beyond the lecture notes. Students suggested several improvements, such as the introduction of milestones to track study progress and the incorporation of various response formats, such as diagrams and tables, to enhance engagement. While GPTutor has proven effective, addressing these weaknesses could further enhance its educational value.

TABLE 4 shows the comparison of the mean exam scores between the seldom-using group (i.e., low-engagement group) and the active group (i.e., high-engagement group). There was a significant difference in mean exam scores (i.e., 51.9 vs. 62.2 respectively), and this difference was statistically significant (*p*-value of 0.00032). Therefore, there was a positive association between the usage of GPTutor and students' academic performance in the course.

TABLE 2. Survey findings of Study 1

Construct	Average (7-point Likert scale)	Standard Deviation
Perceived Ease of Use (EU)	4.96	0.77
Perceived Usefulness (PU)	4.81	1.12
Intention to Use (IU)	4.91	1.27
Enjoyment (EJ)	4.77	1.10

TABLE 3. Differences in mean survey scores between undergraduate and postgraduate students in Study 1

	Undergraduate		Postgraduate		Sig. (2-tailed)
Construct	M	SD	M	SD	
Perceived Ease of Use (EU)	5.15	0.87	4.96	0.69	0.05
Perceived Usefulness (PU)	4.90	1.02	4.81	1.18	1.12
Intention to Use (IU)	4.80	1.07	4.91	1.38	0.09
Enjoyment (EJ)	4.88	1.10	4.77	1.10	0.11

Note. M = Mean. SD = Standard Deviation. Every survey question is constructed with a 7-point Likert scale.

TABLE 4. Differences in mean exam scores between the low- and high-engagement groups in Study 2

Message Range	Student Count	Mean Exam Score	Sig. (2-tailed)
[0, 10)	66	51.90	0.00032
[10, 124]	63	62.18	

V. Summary and Future Research

This paper proposed and demonstrated GPTutor, a GenAI-powered ITS, to facilitate students' learning process. A pilot evaluation study investigated students' perception of using GPTutor for learning and their user experience and engagement in using the platform. The results showed that students generally benefited from learning with GPTutor. Also, some students appreciated that the chatbot can provide a precise summary of lecture content to facilitate course review, while other students suggested that the chatbot should support more in-depth discussion. Finally, the findings revealed a positive association between the usage of GPTutor and students' academic performance.

For future research, the focus is on enhancing the knowledge-based question-answering pipeline, providing better handling of different types of user questions, and incorporating gamification elements to boost student engagement and motivation. It is also scheduled to expand the platform's adoption across more courses to comprehensively evaluate its effectiveness.

Acknowledgment

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